Statistical Methods in Education Research
Vanderbilt University
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48 total points

Problem Set 7 SOLUTIONS

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1. (**7 points**) A random sample of 172 new college graduates was asked to rate on a scale of 1 (not concerned) to 5 (very concerned) their level of concern about meeting future student loan obligations. The sample mean rating was 3.17 and the sample standard deviation was 0.70. Test at the 1% significance level the null hypothesis that the population mean rating is 3.0 against the alternative that it is different from 3.0. Write down the null and alternative hypotheses, and report your test statistic, p-value, and conclusion. (Note: this variable is ordinal-measured and arguably not appropriate for such a test. However, ignore this and treat it as an interval measure here).

   - NOTE: please do this problem "manually" first (show your calculations) then using Stata's `ttesti` calculator.

$$H_0 : \mu = 3.0$$
$$H_1 : \mu \neq 3.0$$

The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{3.17 - 3.00}{0.70/\sqrt{172}} = 3.185$$

In other words, the sample mean of 3.17 is 3.185 standard errors above the hypothesized population mean of 3.0. The probability of obtaining this test statistic or something more extreme in *either direction* if H_0 were true is $p = 2 \times P(t > 3.185) = 2 \times 0.0009 = 0.0017$ (uses $df = 171$). Because $p < \alpha = 0.01$, we reject H_0 in favor of H_1.

See attached log for Stata `ttesti` version.

2. (**7 points**) Children graduating from a particular high school in New York City score an average of 55 on a college-readiness test. A random sample of 36 students from this school is selected to participate in Project Advance, a special program to help high school students prepare for college. At the end of the program, the 36 students are given the college-readiness test. They obtain a sample mean of $\bar{x} = 58.5$ and a sample standard deviation of $s = 12$. Conduct a hypothesis test at significance level $\alpha = 0.05$ to determine whether students who participate in Project Advance score higher on average on the college readiness test than those in the school as a whole. State your null and alternative hypotheses, your test statistic, p-value, and conclusion.

- NOTE: please do this problem "manually" first (show your calculations) then using Stata's `ttesti` calculator.

$$H_0 : \mu = 55$$
$$H_1 : \mu > 55$$

The test statistic is $t = 1.75$ (below). Because s is used instead of σ, we know this test statistic has a t distribution with $n - 1 = 35$ degrees of freedom:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{58.5 - 55}{12/\sqrt{36}} = 1.75$$

From the t-distribution table (or Stata), the p-value is $p = P(t > 1.75) = 0.0446$. Because $p < \alpha$, we reject H_0 in favor of H_1.

See attached log for Stata `ttesti` version.

3. (**9 points**) Same-sex marriage was legalized across Canada by the Civil Marriage Act enacted in 2005. Was this decision supported by a *majority* of the Canadian population? A poll conducted in July 2005 of 1,000 Canadians asked whether this bill should stand or be repealed. The responses were: 55% should stand, 39% should repeal, 6% don't know. Let π represent the population proportion of Canadians who believe it should stand. For testing $H_0 : \pi = 0.50$ against $H_1 : \pi > 0.50$:

   (a) Find the standard error, and interpret in words.

   (b) Find the test statistic, and interpret.

   (c) Find the p-value, and interpret in this context.

   (d) What is your conclusion about H_0?

   (e) Verify your answer using Stata's `ptesti` calculator.

   Note: consider the "don't know" group to be part of the "should repeal" group (i.e., those who *didn't* say the bill should stand.)

The standard error is 0.0158 (below). This is a measure of variability in the sample proportion ($\hat{\pi}$) when the sample size is 1000. Note that the value of π assumed under the null hypothesis is used for the significance test, because we are interested in the sampling distribution of $\hat{\pi}$ when H_0 is true.

$$\sqrt{\frac{\pi_0(1 - \pi_0)}{n}} = \sqrt{\frac{0.50(1 - 0.50)}{1000}} = \mathbf{0.0158}$$

The test statistic is 3.16 (below). If H_0 is true, our sample proportion is 3.16 standard errors above the population mean of 0.50.

$$z = \frac{\hat{\pi} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} = \frac{0.55 - 0.50}{0.0158} = \mathbf{3.16}$$

From the standard normal, $P(z > 3.16) = 0.0008$. If H_0 is true, the probability of obtaining a random sample of 1000 with a sample proportion this far from the population mean would be 0.0008 (i.e., very unlikely). We can use the standard normal *(z)* distribution to find p, since the standard deviation is "known" under the null. (H_0 tells us that $\pi = 0.50$, and that piece of information alone determines the standard deviation of a proportion). Also, the sample size is quite large.

We can **reject** H_0 since $p < \alpha$. (We can assume $\alpha = 0.05$ here but we would reject even under a more stringent α of 0.01).

See attached log for Stata `prtesti` version.

4. (**5 points**) Below are the results of a signifiance test about the population mean self concept score in 8th grade (*slfcnc08*), using a random sample of 25 students. Some information from the table has been removed. Given what remains in the table, what is the standard error of the sample mean? Explain how you arrived at your answer.

```
. ttest slfcnc08=20

One-sample t test
------------------------------------------------------------------------------
Variable |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
slfcnc08 |      25       21.96                              19.62959    24.29041
------------------------------------------------------------------------------
    mean = mean(slfcnc08)                                         t =
Ho: mean = 20                                 degrees of freedom =       24

    Ha: mean < 20               Ha: mean != 20                  Ha: mean > 20
 Pr(T < t) = 0.9523         Pr(|T| > |t|) = 0.0954          Pr(T > t) = 0.0477
```

There are two ways one can find this answer. One is to recognize that the 95% confidence interval is calculated as $\bar{x} \pm t * se(\bar{x})$. The t that would have been used in this CI is $t(df, \alpha/2) = t(24, 0.025) = 2.0639$. With an upper bound of 24.29041 in the CI, this would imply that: $se(\bar{x}) = (24.29041 - 21.96)/2.0639 = \mathbf{1.129}$.

Another way would be to start with the p-value of 0.0477 and to figure out what test statistic it is associated with. Using `display invttail(24,0.0477)`, this t-statistic is 1.7359. This value represents the number of standard errors $\bar{x}$ is from H_0 of 20. Since $1.7359 = (21.96 - 20)/se(\bar{x})$ then $se(x) = (21.96 - 20)/1.7359 =$ **1.129**.

For questions #5-6, use the following Stata dataset from Github:

```
use https://github.com/spcorcor18/LPO-8800/raw/main/data/Grade4_classrooms.dta, clear
```

This file represents a random sample of 4th grade classrooms located in urban school districts in Texas. Each observation is a classroom, and the variables either describe the teacher (e.g., total teaching experience, teacher race/ethnicity), or are an average for the classroom (e.g., math z-scores, % of students who are economically disadvantaged). All data are from 2006.

5. (**14 points: 2 each**) For this problem you will be conducting two-sided tests for the classroom mean math score (*mathz_class*) and mean reading score (*readz_class*) in urban Texas school districts.[1]

   (a) Evaluate the tenability of the normality assumption for these variables or indicate why this assumption is not an issue for this analysis (i.e., that the results of a t-test would not be compromised by a failure to meet this assumption).

   (b) For the state of Texas, the mean classroom math and reading (z-)scores are both 0. Write down the null and alternative hypotheses for t-tests to determine whether the mean urban 4th grade classroom average z-scores for math and reading are zero, or are different from zero.

   (c) Using the appropriate Stata command, what are the test statistics and p-values associated with these tests?

   (d) Use the p-values in each case to determine whether or not H_0 can be rejected in favor of the alternative. Use a significance level of $\alpha = 0.05$.

   (e) Provide 95% confidence intervals for the mean math and reading z-scores in urban Texas classrooms.

   (f) Use the confidence intervals found in part (e) to conduct the tests in parts (b)-(d). Are the results consistent? Why or why not?

   (g) Would you be more or less likely to reject the null hypotheses if the sample size had been 1,000 classrooms, rather than the sample size(s) used above? Explain.

[1] To be clear, these variables are averages of the students' own z-scores in math and reading in a classroom; they are not the z-score for the classroom in the distribution of classrooms. This is not important for the problem, just clarification.

See the attached Stata log.

6. (**6 points**) Use the same data from Question #5 to test whether urban Texas classrooms on average exclude more than 14 percent of their students from state testing because of special education accommodations. (The relevant variable is *excl_sped*).

See attached Stata log. Since *excl_sped* is a proportion, I'll refer to it as π. The hypothesis being tested is:

$H_0 : \pi = 0.14$ and $H_1 : \pi > 0.14$

The t-statistic is 2.8522 with a p-value for this alternative hypothesis of 0.0022. Since $p < \alpha = 0.05$, we can reject at the 95% significance level the null hypothesis in favor of the alternative that the exclusion rate is higher than 14%.

```
. // ************************************************************************
. // LPO.8800 Problem Set 7 - Solutions
. // Last updated: October 20, 2026
. // ************************************************************************
.
. // *************
. // Question 1
. // *************
.
. ttesti 172 3.17 0.70 3

One-sample t test
------------------------------------------------------------------------------
         |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
       x |     172        3.17    .0533745          .7    3.064642    3.275358
------------------------------------------------------------------------------
    mean = mean(x)                                                t =   3.1850
H0: mean = 3                                     Degrees of freedom =      171

    Ha: mean < 3                 Ha: mean != 3                 Ha: mean > 3
 Pr(T < t) = 0.9991         Pr(|T| > |t|) = 0.0017          Pr(T > t) = 0.0009

.
. // *************
. // Question 2
. // *************
.
. ttesti 36 58.5 12 55

One-sample t test
------------------------------------------------------------------------------
         |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
       x |      36        58.5           2          12    54.43978    62.56022
------------------------------------------------------------------------------
    mean = mean(x)                                                t =   1.7500
H0: mean = 55                                    Degrees of freedom =       35

    Ha: mean < 55                Ha: mean != 55                 Ha: mean > 55
 Pr(T < t) = 0.9556         Pr(|T| > |t|) = 0.0889          Pr(T > t) = 0.0444

.
. // *************
. // Question 3
. // *************
.
. prtesti 1000 0.55 0.50

One-sample test of proportion                      x: Number of obs =     1000
------------------------------------------------------------------------------
             |       Mean   Std. err.                      [95% conf. interval]
-------------+----------------------------------------------------------------
           x |        .55   .0157321                      .5191656    .5808344
------------------------------------------------------------------------------
    p = proportion(x)                                             z =   3.1623
H0: p = 0.5

     Ha: p < 0.5                 Ha: p != 0.5                  Ha: p > 0.5
 Pr(Z < z) = 0.9992         Pr(|Z| > |z|) = 0.0016          Pr(Z > z) = 0.0008
```

```
.
. // *************
. // Question 5
. // *************
.
. use https://github.com/spcorcor18/LPO-8800/raw/main/data/Grade4_classrooms.dta
> , clear

.
. // ********
. // Part a
. // ********
. histogram mathz_class, name(hist1, replace) nodraw
(bin=28, start=-3.0838358, width=.15680618)

. histogram readz_class, name(hist2, replace) nodraw
(bin=27, start=-4.6651998, width=.23412534)

.
. graph combine hist1 hist2 , xsize(3) ysize(6) col(1)

. graph export hists.pdf, as(pdf) replace
file hists.pdf saved as PDF format

.
. // Both distributions are left skewed, and the math distribution has a mass of
. // observations that are especially low (around -3). Because our sample size i
> s
. // large, the CLT predicts that means calculated from samples drawn of this si
> ze
. // will be approximately normally distributed. There is a greater risk of
. // departure from normality for math vs. reading, so the mass of low
. // observations are worth further investigation.
.
. // **********
. // Parts b-d
. // **********
. // H0: mu_m = 0 and H1: mu_m neq 0
. // H0: mu_r = 0 and H1: mu_r neq 0
.
. // Results shown below. The t-statistic for math is -5.2303. For reading, it i
> s
. // -1.2237. The p-values for a two-sided hypothesis test are <0.0001 and 0.221
> 5,
. // respectively.
.
. ttest mathz_class=0

One-sample t test
------------------------------------------------------------------------------
Variable |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
mathz_~s |     816   -.1590247    .0304047     .868533   -.2187056   -.0993439
------------------------------------------------------------------------------
    mean = mean(mathz_class)                                      t =  -5.2303
H0: mean = 0                                     Degrees of freedom =      815

    Ha: mean < 0                 Ha: mean != 0                 Ha: mean > 0
 Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000
```

```
. ttest readz_class=0

One-sample t test
------------------------------------------------------------------------------
Variable |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
readz_~s |     765   -.0255273    .0208613    .5769953   -.0664796     .015425
------------------------------------------------------------------------------
    mean = mean(readz_class)                                      t =  -1.2237
H0: mean = 0                                     Degrees of freedom =      764

    Ha: mean < 0                 Ha: mean != 0                 Ha: mean > 0
 Pr(T < t) = 0.1107         Pr(|T| > |t|) = 0.2215          Pr(T > t) = 0.8893

.
. // Because p<alpha for math, we reject the null hypothesis that the mean
. // classroom math score in urban Texas districts is the same as the statewide
. // average of zero. As p>alpha for reading, we do not reject the null hypothes
> is
. // that the mean classroom reading score in urban Texas districts is the same
> as
. // the statewide average of zero. (It is notable, however, that there are a
. // number of missing values for readz_class. One reason this may be is that
. // some classrooms are entirely English language learners who are exempted
. // from the reading test).
.
. // **********
. // Parts e-f
. // **********
. // 95% confidence intervals
.
. // The 95% confidence intervals are provided in the Stata output below:
. // (-0.219, -0.099) for math, and (-0.066, 0.0015) for reading.
.
. // The mean under the null hypothesis (0) is outside the 95% confidence
. // interval for math, so we can reject that null hypothesis. Zero is contained
>
. // within the 95% confidence interval for reading, so we cannot reject that
. // null hypothesis. These are the same conclusions reached using the p-value
. // method.
.
. mean mathz_class

Mean estimation                           Number of obs = 816

--------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+------------------------------------------------
 mathz_class |  -.1590247   .0304047     -.2187056   -.0993439
--------------------------------------------------------------

. mean readz_class

Mean estimation                           Number of obs = 765

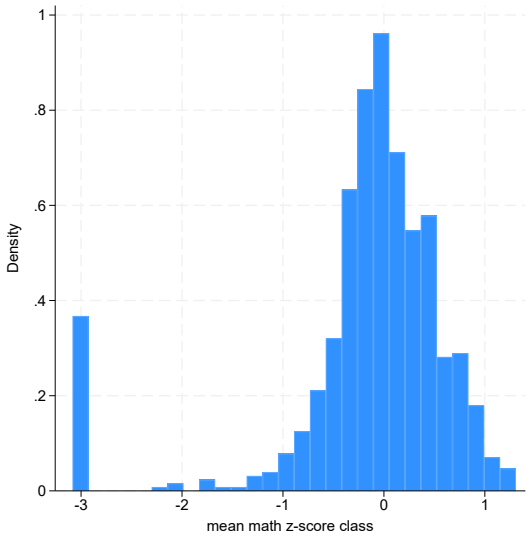
--------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+------------------------------------------------
 readz_class |  -.0255273   .0208613     -.0664796     .015425
--------------------------------------------------------------
```

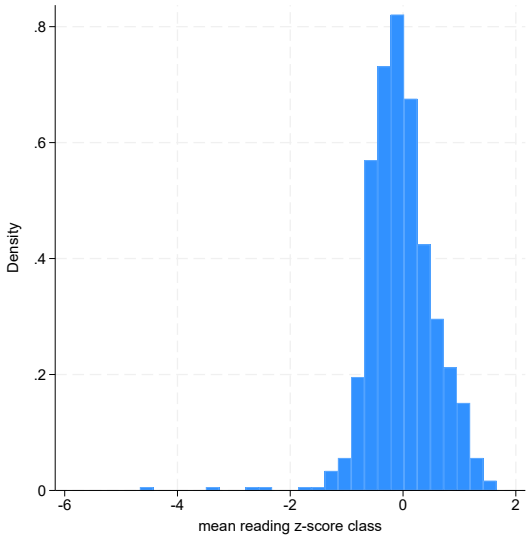
```
.
. // **********
. // Part g
. // **********
.
. // You would be more likely to reject the null hypothesis with a larger sample
>
. // size. All else equal, a larger n is associated with a smaller p-value.
. // Another way to think about this is that the confidence interval around x-ba
> r
. // will be narrower, making it less likely that zero will be contained within
> it.
.
.
.
. // *************
. // Question 8
. // *************
.
. ttest excl_sped=0.14

One-sample t test
------------------------------------------------------------------------------
Variable |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
excl_s~d |     835    .1659154     .009086    .2625515    .1480813    .1837494
------------------------------------------------------------------------------
    mean = mean(excl_sped)                                        t =   2.8522
H0: mean = 0.14                                 Degrees of freedom =      834

   Ha: mean < 0.14             Ha: mean != 0.14               Ha: mean > 0.14
 Pr(T < t) = 0.9978         Pr(|T| > |t|) = 0.0044          Pr(T > t) = 0.0022

.
.
. // Close log and convert to PDF
.         capture log close
```


Density
0
.2
.4
.6
.8
1
-3
-2
-1
0
1
mean math z-score class


Density
0
.2
.4
.6
.8
-6
-4
-2
0
2
mean reading z-score class